claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-aa1a0efb622a445ad.jsonl`
- SHA-256: `b6dcc1b8db27e3fb2bdddf24d335abb18f5ef903a9effd757f0b2f8b4e32fc95`
- Source modified: `2026-06-06T17:30:26+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:28:43.705Z)

Web Searcher: classical/lightweight

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Your angle: **classical/lightweight** ? Covers cheap, CPU-friendly classical signal methods (change-point / HMM on the trajectory the user already has) plus scoreboard OCR as a low-compute precision source for point boundaries.
Search query: `change point detection HMM shuttle trajectory rally segmentation scoreboard score overlay OCR point boundary sports video`

Task
Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.
Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.
Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-06T17:28:47.226Z)

I'll research this. Let me start with the targeted search on classical/lightweight methods.

3. assistant (2026-06-06T17:28:50.109Z)

I have the search tool. Let me run several targeted searches in parallel to cover change-point detection / HMM on trajectory and scoreboard OCR for point boundaries.

4. user (2026-06-06T17:28:57.961Z)

Web search results for query: "change point detection HMM shuttle trajectory rally segmentation badminton video"

Links: [{"title":"Future Prediction of Shuttlecock Trajectory in Badminton Using Player?s Information","url":"https://www.mdpi.com/2313-433X/9/5/99"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"Badminton Video Analysis based on Spatiotemporal and Stroke Features","url":"https://www.researchgate.net/publication/317159692_Badminton_Video_Analysis_based_on_Spatiotemporal_and_Stroke_Features"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | Request PDF","url":"https://www.researchgate.net/publication/362900461_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"},{"title":"[2204.01899] MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://ar5iv.labs.arxiv.org/html/2204.01899"},{"title":"Prediction of Shuttle Trajectory in Badminton Using Player?s Position","url":"https://www.scitepress.org/PublishedPapers/2023/117858/117858.pdf"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/pdf/2307.16000"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}]

The search results provide relevant research on badminton video analysis, shuttle trajectory reconstruction, and rally segmentation. Here are the key findings:

Rally Segmentation and Shot Detection

Shot segmentation is critical for reconstructing 3D trajectories, with a shot occurring when a player hits the shuttle with their racket and ending right before the opposing player hits it or when the shuttle hits the court, making shot identification equivalent to identifying hit events.

Badminton videos generally contain multiple rallies, and each rally contains multiple shots, with segmentation being challenging due to the extreme speed of the shuttlecock.

Hit Detection Methods

Hit detection can be formulated as a multi-class classification problem, with three classes: *no hit*, *near player hit*, and *far player hit* predicted at each frame.

One approach uses a GRU-based recurrent network that consumes court, pose, and 2D shuttlecock information to make hit predictions.

Another algorithm iterates over each shuttlecock flying direction in the input sequence and checks for specific transitions to detect hit-frames when the shuttlecock's flying direction has changed.

Trajectory Reconstruction

Automating rally segmentation and shuttlecock position detection in match videos would be useful for practical applications, and systems can detect the trajectory of the shuttlecock, thereby enabling identification of players' swing moments and segmentation of each round.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-06T17:29:10.197Z)

Web search results for query: "scoreboard score overlay OCR point boundary detection sports broadcast video segmentation"

Links: [{"title":"Identification of replay segments","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/7653131"},{"title":"Identification of replay segments","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/7474698"},{"title":"(PDF) A Real-Time Score Detection and Recognition Approach for Broadcast Basketball Video","url":"https://www.researchgate.net/publication/221262832_A_Real-Time_Score_Detection_and_Recognition_Approach_for_Broadcast_Basketball_Video"},{"title":"Scoreboard OCR Meets Logo Metrics: Context Matters","url":"https://medium.com/@API4AI/scoreboard-ocr-meets-logo-metrics-context-matters-14824e6c9325"},{"title":"Automated Wicket-Taking Delivery Segmentation and Trajectory-Based Dismissal-Zone Analysis in Cricket Videos Using OCR-Guided YOLOv8","url":"https://arxiv.org/pdf/2510.18405"},{"title":"Automated Wicket-Taking Delivery Segmentation and Trajectory-Based Dismissal-Zone Analysis in Cricket Videos Using OCR-Guided YOLOv8","url":"https://arxiv.org/html/2510.18405"},{"title":"Free - ScoreSight - Free Open Source OCR tool for Gaming and Scoreboards | OBS Forums","url":"https://obsproject.com/forum/resources/scoresight-free-open-source-ocr-tool-for-gaming-and-scoreboards.1884/"},{"title":"Methods and systems for scoreboard text region detection","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11010627"}]

Based on the search results, here's what I found regarding scoreboard score overlay OCR and point boundary detection in sports broadcast video segmentation:

Scoreboard Detection and OCR in Sports Broadcasts

Scoreboard OCR is a camera-based way to read the scoreboard directly from the broadcast video ? the game clock, score, period/half/inning, and status flags like penalties or power plays.

Video Segmentation Applications

The existence of a scoreboard in the video provides an indication of an associated replay segment and potentially an interesting event, with scoreboards being overlaid on frames of football and baseball broadcast video. Scoreboards are not overlaid on frames during replays, which can be used to determine or verify the identification of replay segments.

Detection Techniques

The scoreboard may be detected by considering unique features such as: there being only one scoreboard in a frame, a scoreboard being at specific locations depending on the sport, most frames being dominated by the court or field, and a scoreboard appearing in frames of "plays" where the game is ongoing.

Sport-Specific Applications

Score overlays provide compact and continuously updated contextual information, serving as useful cues for video indexing and alignment. Research has shown that on-screen score information can act as a stable semantic anchor in sports videos using scoreboard localization and recognition frameworks based on SIFT matching and OCR enhancement.

Text Region Detection Challenges

OCR processes applied to video frames can result in recognizing numerous characters unrelated to desired sport information, including text on advertisements and spectators' clothing, requiring additional processes that can be inefficient and error-prone.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:29:13.760Z)

Good results. Let me run two more searches to cover classical change-point/HMM in tennis/table-tennis rally segmentation and the ShuttleSet dataset licensing, plus the cricket OCR-guided segmentation paper which is directly relevant.

7. user (2026-06-06T17:29:21.032Z)

Web search results for query: "ShuttleSet badminton stroke dataset license commercial use TrackNetV2 dataset license"

Links: [{"title":"SENTATION FUSION NETWORK FOR PRE-STROKE PRE","url":"https://openreview.net/pdf/e360f0b2cb53ae8733ca7f6703925a362e9d94bd.pdf"},{"title":"(PDF) ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis","url":"https://www.researchgate.net/publication/371413847_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining","url":"https://dl.acm.org/doi/10.1145/3580305.3599906"},{"title":"[PDF] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Semantic Scholar","url":"https://www.semanticscholar.org/paper/ShuttleSet:-A-Human-Annotated-Stroke-Level-Singles-Wang-Huang/405132e19aa07c8a9ae6f8a4e6f440b00a105027"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"(PDF) ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset","url":"https://www.researchgate.net/publication/371909519_ShuttleSet22_Benchmarking_Stroke_Forecasting_with_Stroke-Level_Badminton_Dataset"},{"title":"[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis","url":"https://arxiv.org/abs/2306.04948"},{"title":"ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset","url":"https://arxiv.org/html/2306.15664v1"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton   Tactical Analysis","url":"https://arxiv.org/pdf/2306.04948"}]

Based on the search results, I found information about both datasets, though the results don't explicitly detail the specific commercial use licenses. Here's what I found:

ShuttleSet

ShuttleSet is the largest publicly-available badminton singles dataset with annotated stroke-level records, containing 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021. ShuttleSet is manually annotated with a computer-aided labeling tool to increase the labeling efficiency and effectiveness of selecting the shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke.

The search results show that baseline codes and the dataset are made available on GitHub, suggesting an academic/open-source context, but the specific license terms for commercial use are not detailed in these results.

TrackNetV2

The TrackNetV2 dataset is a public dataset containing 77k annotated frames from 26 unique singles matches from international play, filmed from an overhead, static, broadcast-view camera.

Unfortunately, the search results do not contain specific information about the commercial use licenses for either dataset. For detailed license information, I would recommend checking the official GitHub repositories or the dataset documentation directly, as these details would typically be specified in a LICENSE file or README documentation.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-06T17:29:22.566Z)

Web search results for query: "rally play break segmentation tennis table tennis classical signal trajectory features audio hit detection lightweight"

Links: [{"title":"(PDF) Detection of ball hits in a tennis game using audio and visual information","url":"https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"},{"title":"(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis","url":"https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"},{"title":"Table Tennis Stroke Detection and Recognition Using Ball Trajectory Data | Request PDF","url":"https://www.researchgate.net/publication/362039797_Table_Tennis_Stroke_Detection_and_Recognition_Using_Ball_Trajectory_Data"},{"title":"TennisExpert: Towards Expert-Level Analytical Sports Video Understanding","url":"https://arxiv.org/pdf/2603.13397"},{"title":"TT3D: Table Tennis 3D Reconstruction","url":"https://arxiv.org/pdf/2504.10035"},{"title":"Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/"},{"title":"Egocentric Event-Based Vision for Ping Pong Ball Trajectory Prediction","url":"https://arxiv.org/pdf/2506.07860"},{"title":"Spin Detection Using Racket Bounce Sounds in Table Tennis This research was funded by Sony AI.","url":"https://arxiv.org/html/2409.11760v1"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton   video","url":"https://arxiv.org/pdf/2204.01899"}]

Based on the search results, I found relevant research on rally play break segmentation, signal trajectory features, and audio hit detection in tennis and table tennis:

Rally and Break Segmentation

Professional tennis broadcasts typically consist of alternating periods of rally play and pauses between points, and to organize datasets at the rally level, full match broadcasts are segmented into individual rally clips.

Audio-Based Hit Detection

A lightweight, purely audio-based detection model can be employed by training a neural network to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track. To ensure the model is robust against complex ambient stadium noises, training data can be augmented by overlaying various background sounds such as crowd cheering, clapping, and announcer voices onto audio samples.

Ball Trajectory Segmentation

To analyze individual ball trajectories, rallies can be segmented using piecewise polynomial segmentation, since a ball's flight trajectory can be well approximated by a second-degree polynomial, which naturally aligns with the motion characteristics of the ball.

Table Tennis-Specific Applications

A scheme combining audio and visual information can be proposed for structure indexing and highlight generating, where by taking advantage of temporal relationship between audio and visual signals, scene clusters can be specified with semantic labels including rally scenes and break scenes.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:29:35.246Z)

I have strong coverage. Let me verify the most directly relevant classical/lightweight result ? the hit-frame detection paper (trajectory direction change, no deep model needed) and the cricket OCR-guided segmentation paper ? to confirm their methods before ranking.

10. user (2026-06-06T17:29:41.847Z)

I've received what appears to be a PDF file rendered as binary data rather than readable text. The content provided consists primarily of compressed/encoded streams (FlateDecode format) that cannot be meaningfully analyzed for information about hit-frame detection methods.

To provide an accurate summary regarding:
- Classical/rule-based approaches for shuttlecock detection
- Features and accuracy metrics
- Dataset information (ShuttleSet/TrackNet)
- License details

I would need the document in a readable text format (such as plain text, markdown, or properly extracted PDF text).

Could you please provide:
1. The document title and source
2. A text version of the relevant sections, or
3. Re-upload in an accessible format

This will allow me to deliver the specific technical details you're requesting while maintaining the character limits and citation standards you've outlined.

[Binary content (application/pdf, 1.5MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780766977975-dnjzkl.pdf]

11. user (2026-06-06T17:29:44.859Z)

OCR-Guided Temporal Segmentation in Cricket Video Analysis

How Scoreboard OCR Guides Segmentation

The system uses optical character recognition on broadcast scorecard overlays to automatically identify wicket events. The process works by:

1. **Frame sampling** at fixed intervals (0.1s default) and extracting a predefined scorecard region of interest
2. **Preprocessing** the scorecard image through grayscale conversion, power transformation (?=7), intensity inversion, and morphological operations to enhance text readability
3. **Score parsing** using OCR followed by validation against common score formats (Wicket-Runs, Wicket/Runs, etc.)

Boundary Signal Detection

"A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance. Once such an increment is detected, the corresponding delivery segment is extracted and saved for downstream spatial analysis."

This wicket count transition serves as the temporal trigger for isolating relevant video portions rather than processing entire matches uniformly.

OCR Tool and Performance

The paper does **not specify which OCR engine** was used, nor does it provide **computational cost metrics**.

Regarding accuracy, the authors note that the OCR stage "was effective in identifying wicket-taking delivery segments across multiple match videos," but acknowledge limitations: "segmentation quality may vary when the broadcast overlay is partially occluded, reformatted, or temporarily absent."

No quantitative accuracy numbers are reported for the OCR or segmentation stages?only qualitative assessment.

12. assistant (2026-06-06T17:29:48.939Z)

The cricket OCR paper confirms the scoreboard-score-change-as-boundary mechanism well. The hit-frame PDF didn't parse, but I have enough from the search snippet (it includes a rule-based algorithm iterating over shuttlecock flying direction and detecting transitions ? a classical signal method). I have solid coverage across both angles. Let me compile the ranked results.

13. user (2026-06-06T17:30:12.225Z)

Structured output provided successfully

14. assistant (2026-06-06T17:30:26.392Z)

Returned 6 ranked results via StructuredOutput. Summary of what I found on the classical/lightweight angle:

Top hits (high relevance):
- **Automated Hit-frame Detection (arXiv 2307.16000)** ? the most on-substrate result: a classical, rule-based change-point heuristic on the shuttle flying-direction (CPU-cheap, no deep model) that fires on direction reversals, plus a GRU fusion option. Hit-frames are the within-rally shot boundaries the user can cluster into rallies.
- **OCR-Guided Cricket Segmentation (arXiv 2510.18405)** ? the cleanest concrete blueprint for "score-change = point boundary": fixed-interval frame sampling, fixed scorecard ROI, lightweight preprocessing, OCR, and a boundary trigger on score increment. Honest about occlusion/absent-overlay failure modes, so it's a precision-complement, not a standalone.
- **Tennis ball-hit detection via audio+visual** ? validates the locally-runnable fuse-audio-thwack-with-trajectory lever to attack WASB's recall bottleneck.

Supporting (medium):
- **Shuttle tracking + hit-detection fusion (PMC11244353)** ? badminton-specific, trajectory-only refinement of shot boundaries.
- **ShuttleSet (KDD 2023)** ? ready-made rally/stroke labels for a golden eval set, with a license caveat flagged (no stated permissive commercial license; pro-broadcast footage ? must verify LICENSE before commercial training).
- **Table-tennis audio-only hit detection** ? transferable pure-audio rally-vs-break clustering recipe.

Sources:
- [Automated Hit-frame Detection for Badminton Match Analysis](https://arxiv.org/pdf/2307.16000)
- [OCR-Guided Wicket-Taking Delivery Segmentation in Cricket Video](https://arxiv.org/html/2510.18405)
- [Detection of Ball Hits in a Tennis Game Using Audio and Visual Information](https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information)
- [Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection (PMC11244353)](https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/)
- [ShuttleSet (KDD 2023)](https://arxiv.org/abs/2306.04948)
- [Ball Hit Detection in Table Tennis Games Based on Audio Analysis](https://www.researchgate.ne

t/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis)